

Georges Parfait Djimefo Kapen | AI Developer | Agentic AI & Cloud Systems

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Profile

AI developer with hands-on experience designing and deploying **production-grade agentic AI systems**, **multi-agent LLM workflows**, and **enterprise-scale RAG chatbots** on AWS. Strong background in **AI evaluation & benchmarking**, **cloud-native architecture** (serverless, containers), **CI/CD automation**, and **ETL/ELT data pipelines**. PhD expected Summer 2026 with research in **AI**, **graph neural networks**, **multi-agent reinforcement learning**, and **cybersecurity**. Proven ability to communicate technical trade-offs to executives and non-technical stakeholders. Innovation, teamwork, leadership, and results-oriented.

Technical Skills

LLM / Agents: LangChain, LangGraph, LangSmith, MCP, RAG (FAISS), CRAG, RAGAS, Claude (Haiku, Opus), Kiro
AI / ML: PyTorch, TensorFlow, Scikit-Learn, MARL (QMIX, QTRAN), GNN, DQN, NLP, Computer Vision, LLMs
Cloud: AWS (SageMaker, Bedrock, EKS, Lambda, CloudWatch, S3, ECR, CloudFormation), OpenStack, Kubernetes, Docker
DevOps / IaC: CI/CD (GitHub Actions, CodeBuild, OIDC), Git, Jenkins, GitLab, Terraform, Infrastructure as Code
Data: PySpark, Databricks, Snowflake, Delta Lake, SQL, Relational Databases, ETL/ELT
Languages: Python, Java, C++, PHP, SQL, R, JavaScript, TypeScript
Web / APIs: REST APIs, FastAPI, Streamlit, React, Angular, Node.js, D3.js, Plotly, Dash, Power BI, Tableau
Methods: Agile/Scrum, TDD, Design Patterns, Statistical Modeling, MLOps, AI Governance

Relevant Experience

Ericsson

Montréal

ML Intern — Agentic AI & Cloud Systems

2026

- Designed and deployed a **production RAG Autonomous Agentic AI system** that automates telecom radio unit log anomaly detection, replacing manual review for enterprise clients.
- Built a **multi-agent LLM workflow**: 6-tool ReAct agent orchestrated by Claude Haiku 4.5 on Bedrock (SageMaker scoring, FAISS RAG, RF physics world model, pattern classification, CloudWatch health monitoring), replicating analyst reasoning.
- Evaluated and benchmarked foundation models with **RAGAS** faithfulness/relevancy scoring and evidence-based **meta-cognition** confidence grading (HIGH/MEDIUM/LOW) to optimize performance-cost trade-offs.
- Trained a custom **Dual-Attention SLM** achieving **F1 90.6%**, 99.97% accuracy, and **1 ms/log** inference latency on AWS SageMaker.
- Engineered a **production-scale map-reduce pipeline**: triage → cluster by failure signature → parallel agent investigation → adaptive cross-cluster synthesis. Handles 120+ anomalies in ~12 min.
- Built an **executive-facing Streamlit dashboard** exposing accuracy, usage, latency, and cost-per-query metrics for leadership decisions.
- Deployed as a **K8s pod on EKS** via Cloudflare tunnel; built an **event-driven pipeline** (S3 → Lambda → SQS → K8s worker → S3 → SNS alerts) for high availability.
- Designed a **fully autonomous retraining loop**: drift-triggered + weekly via EventBridge, with auto-promotion only when new model F1 exceeds production baseline (AI governance & monitoring).
- Implemented a **7-stage CI/CD pipeline** (GitHub Actions + CodeBuild with OIDC): test → OWASP security scan → CloudFormation infra → staging gate → deploy → smoke test → auto-rollback. Zero static IAM keys.
- Authored **131 automated tests** with an AST-based adaptive test guardian ensuring 100% production function coverage.
- *Stack:* Python, AWS (SageMaker, Bedrock, EKS, Lambda, SQS, SNS, S3, ECR, CloudFormation, CloudWatch), LangChain, LangGraph, LangSmith, FAISS, MCP, Claude, Kiro, Docker, Kubernetes, CI/CD, Terraform.

Intact Financial Corporation

Montréal

AI Developer I — Enterprise Data Systems

2022–2024

- Ingest and process large-scale data.
- Work in an insurance environment.
- Monitor pipelines and develop automated testing solutions ensuring data quality and system reliability.
- *Stack:* PySpark, SQL, Databricks, Snowflake, AWS, Shell, Git.

Additional Experience

Polytechnique Montréal

Montréal

Lecturer

2025–2026

- Teach the Python procedural programming course; communicate complex technical concepts to non-technical audiences.

Polytechnique Montréal

Montréal

Lecturer

2022–2023

- Teach the Python procedural programming course.

Université de Montréal <i>Scientific Assistant — MOOC AI</i>	Montréal 2021–2022
Design courses and practical work in supervised learning (Python).	
Ministry of Tourism and Recreation <i>Contractual Statistical Economist Engineer</i>	Cameroon 2012–2014
Analyze data	
Protestant University of Central Africa <i>Associate Professor of Statistics & Probability</i>	Cameroon 2010–2012
Teach statistics and probability courses.	
ISSEA (Sub-regional Institute of Statistics and Applied Economics) <i>Part-time Lecturer in Data Analysis</i>	Cameroon 2009–2010
Teach the Data Analysis course (R).	

Publications

- [1] *Energy-Efficient and Near Real-Time Routing Algorithm for Disaster Monitoring in Wireless Sensor Networks*
G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal (Submitted)
- Designed **GAPF**, a novel **meta-learning** framework fusing MARL policies (QMIX & QTRAN) via a **GCN encoder** with only **1 473 parameters**.
 - Achieved $\geq 98\%$ packet delivery, $2\text{--}3.5\times$ less energy, $13\times$ less memory than baselines.
- [2] *Carbon-Aware Autonomous Data Reduction and Self-Healing Routing for 6G-Integrated Disaster Sensor Networks*
G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal (Submitted)
- Proposed **CALASH**, the **first protocol** jointly optimizing Lifecycle Carbon Intensity via a hybrid **Lyapunov–DQN** engine.
 - Achieved 60% longer network lifetime and 33% lower LCI than baselines.
- [3] *Zero-Trust Shielded Graph-Structured Decision Control for Secure Autonomous Recovery in AI-Native 6G Networks*
G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal (Submitted)
- Designed **CASTER-ZT**, combining **GNN-based** decision making with a formally bounded **zero-trust** admissibility shield; enterprise security and compliance by design.
 - Achieved 74–98% rogue detection with **zero false positives**, 3 ms latency within O-RAN 10 ms budget.

Selected Projects

Intact Financial Corporation <i>Data Engineering Intern</i>	Montréal 2022
Manipulate data (PySpark, SQL, Databricks, Snowflake, AWS).	
Polytechnique Montréal <i>Integrator Project — IATA</i>	Montréal 2021
“Flight Data eXchange — Develop an initial and contextual flight data assessment tool.” (Dash, Python, React, Electron, Axios).	
Polytechnique Montréal <i>Undergraduate Research Initiation Intern</i>	Montréal 2020–2021
Voice Recognition modeling for secure bank payments; Deep Learning (Python) deployed on Android (Java).	
Polytechnique Montréal <i>NLP Project</i>	Montréal 2021
Extraction of keywords from texts as well as their type (Python).	
Polytechnique Montréal <i>Data Visualization Project — Chaire de recherche en fiscalité</i>	Montréal 2021
Web application to compare taxes and rates by year (Plotly.js, D3).	
Polytechnique Montréal <i>Entrepreneur / Research Initiation Intern</i>	Montréal 2020
Face Recognition modeling integrating racial and gender bias; Deep Learning (Python).	
FLEX GROUP <i>Programming Intern</i>	Laval 2019–2020
Modeling of Automatic Optical Character Recognition (Python, PHP).	
International Bank of Cameroon for Savings and Credit <i>Intern for Master's in Applied Statistics</i>	Cameroon 2006
Mathematical models for forecasting a bank's income and expenses (R).	

Education

Polytechnique Montréal <i>PhD in Computer Engineering</i>	Montréal, Canada Summer 2026
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- AI-driven routing, carbon-aware networking, and zero-trust security for 6G sensor networks.

Polytechnique Montréal

Montréal, Canada

Engineering Diploma in Software Engineering

- Concentration: Artificial Intelligence — Data Sciences.

Advanced National School of Engineering (ENSPY)

Yaoundé, Cameroon

Master's Degree in Applied Statistics

- Equivalency: Master's in Probability — Statistics.

University of Yaoundé I

Yaoundé, Cameroon

Master's Degree in Mathematics

- Equivalency: Specialized Higher Studies Diploma (DESS) in Mathematics.

Licenses & Certifications

- AWS Certified Cloud Practitioner (CLF-C01)
Certification Prep Path (Pluralsight), Montréal
- AWS Certified Developer — Associate (DVA-C02)
Certification Prep Path (Pluralsight), Montréal

Conferences & Seminars

PolyCongrès 2026, Polytechnique Montréal

Montréal, Canada

Oral Presentation

Mar–Apr 2026

"Energy-Efficient and Near Real-Time Routing for Disaster Monitoring in Wireless Sensor Networks — Graph-Attentive Policy Fusion (GAPF)."

20th Conf. African Assoc. of Insect Scientists

Yaoundé, Cameroon

Oral Presentation

Oct 2013

"Biogeography of communities of insects using clustering methods."

29th Annual Conf. Int. Society for Clinical Biostatistics

Copenhagen, Denmark

Poster

Aug 2008

"HIV incidence estimation from repeated cross-sectional surveys."

ANMSA (African Network of Mathematical Statistics and its Applications)

Franceville, Gabon

Oral Presentation

Jan 2008

"HIV epidemiology in Cameroon: Estimating incidence from prevalence data."

Volunteering

- Volunteer at the IEEE branch of Polytechnique Montréal, Montréal, 2018.
- Volunteer at the ConFoo conference, Montréal, 2018.